A CS Guide to Linear Algebra from Scratch

Strings

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Abstract

This guide was made to help programmers learn linear algebra concepts with a CS perspective with as little background knowledge as possible. The main focus will be on understanding the underlying principles behind practical applications as well as how they interact with the theoretical ones.

1 Introduction

Linear algebra is a section of math dedicated to solving systems of linear equations and understanding geometric concepts such as planes and lines. The underlying foundation of linear algebra is vectors and matrices (the plural of "matrix"). From a CS perspective the use of a linear algebra can often be seen in graphics, machine learning and cryptography. This paper will be used to build from the basic fundamentals to an above surface knowledge of the topic from the math and its computational implementation.

2 Basic Concepts

2.1 Scalar

A scalar refers to a normal number used in a vector operation. They are typically **real numbers ($\mathbb{R}$)** or **complex numbers ($\mathbb{C}$)**

2.2 Vectors

Vectors are one of the biggest building blocks for linear algebra. At its core, a vector is just a list of numbers. For example, consider the vector:

$$\mathbf{v} = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} \tag{1}$$

This vector has 3 components: 2, 5, -1.
Vectors are used to store information, a common use of a vector is store location.

2.2.1 Vector Sets

Before we continue lets explain what a **vector set** is. A **vector set** is a collection of vectors, a bundle of vectors put in set. You think of them as "the vectors we're allowed to consider" **Example:**

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\} \tag{2}$$

Here, the set S contains three vectors in 2D space. Anything not in S is not considered part of this particular set.

Later, we'll see a **vector space**, which is a special type of vector set. In a vector space, you can combine vectors or stretch them, and you'll never leave the set — everything still stays inside the set.

2.2.2 Vector Space

Now that we understand what a **vector set** is — a collection of vectors let's talk about a **vector space** It is a special type of vector set where you can do two things freely:

1. **Add any two vectors** from the set, and the result is still in the set.
2. **Multiply any vector by a number** (scalar), and the result is still in the set.

In other words, a vector space is a "magic basket" of vectors: - You can combine vectors or stretch/shrink them. - No matter what you do, you never leave the basket.

Example: All 2D vectors form a vector space:

$$\mathbb{R}^2 = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \middle| \, x, y \in \mathbb{R} \right\} \tag{3}$$

In non-klingon this means: A 2D vector contains **x** and **y**.
As well, as **x** and **y** are **real numbers**.

- Add two vectors:

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix} \in \mathbb{R}^2 \tag{4}$$

- Multiply a vector by a scalar:

$$2 \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix} \in \mathbb{R}^2 \tag{5}$$

Notice how the results stay inside the set — that's the defining feature of a vector space.

2.2.3 Algebraic

There are multiple ways to view a vector in linear algebra, first is algebraically. Algebraically a vector is a element inside a **vector space**.
For example, all 3D vectors form $\mathbb{R}^3$

- You can add component-wise:

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix} \tag{6}$$

- And you can multiply by a scalar:

$$2 \cdot \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 6 \\ -2 \end{bmatrix} \tag{7}$$

2.2.4 Geometric

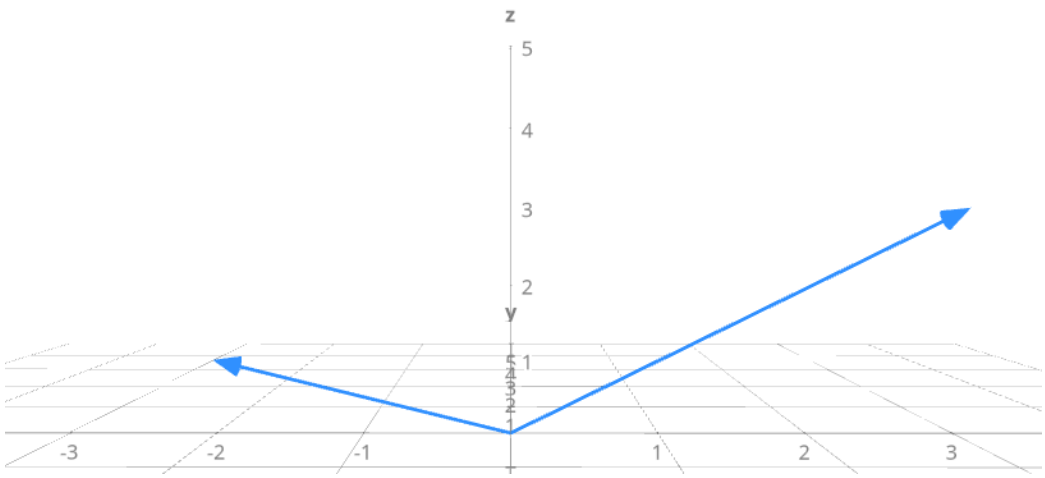

Figure 1: Geometric representation of 2D Vector.

In a 2D Vector, you can visualize a vector as an **arrow** pointing from the origin of **(0, 0)** to a point (x, y).
Example:

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ 3 \end{bmatrix} \mathbf{v}_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \tag{8}$$

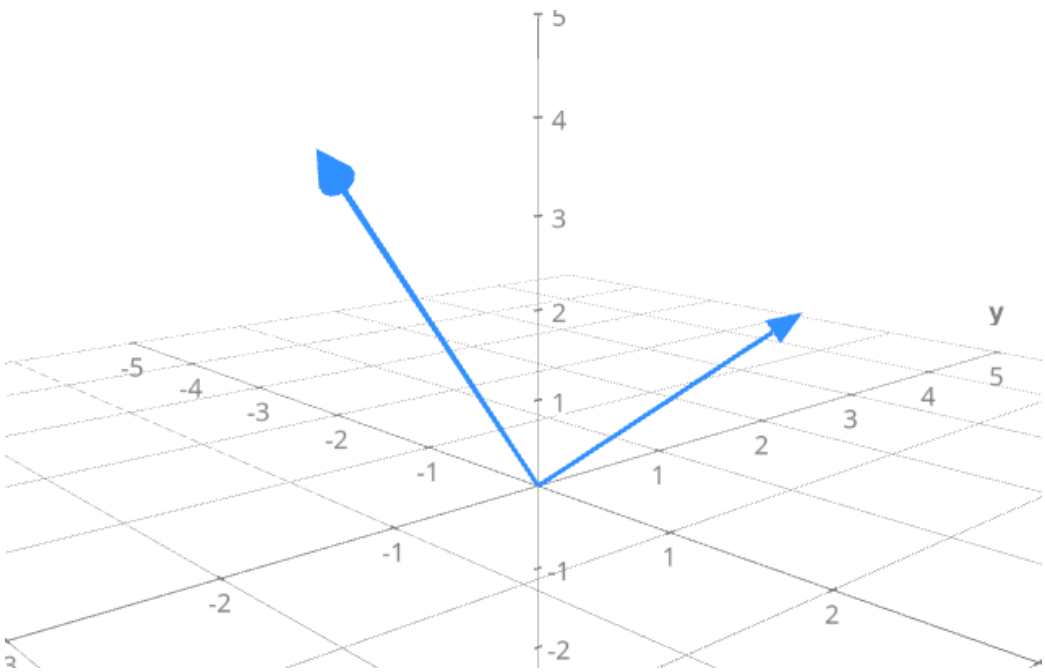

Figure 2: Geometric representation of 3D Vector.

The same principle works with a 3D vector but pointing to (x, y, z)
Example:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} \tag{9}$$

2.2.5 Computer Science Usage

In Computer Science common usages of vectors can be:

- A list/array
- Method to represent states (e.g., pixel colors, probability, features in ML)
- A direction in computer graphics/physics
- A velocity or force
- Rotation

Example 2.1 (C++ Representation)**.** *Here is a example script*

upquote

```cpp
#include <iostream>
#include <vector>

int main() {
    std::vector<int> v1 = {3, 4};
    std::vector<int> v2 = {-2, 1};

    std::vector<int> sum(2);
    for (size_t i = 0; i < 2; ++i) {
        sum[i] = v1[i] + v2[i];
    }

    std::cout << "v1 + v2 = [";
    for (size_t i = 0; i < 2; ++i) {
        std::cout << sum[i];
        f (i < 1) std::cout << ", ";
    }
    std::cout << "]" << std::endl;
    return 0;
}

```

2.2.6 Magnitude

The magnitude is a property of a vector that can be written with "$\|$" surrounding a vector, it measures the length (or norm) of a vector from the Pythagorean theorem:

$$||\mathbf{v}|| = \sqrt{v_1^2 + v_2^2 \cdots + v_n^2} \tag{10}$$

Example:

$$|| \begin{bmatrix} 3 \\ 4 \end{bmatrix} || = \sqrt{3^2 + 4^2} = 5 \tag{11}$$

Note: The subscript **'n'** is the total amount of **components** in a column when discussing vectors.
Also worth noting: When the magnitude of a vector is **equal to 1** it is known as a **unit vector**.

2.3 Vector Operations

2.3.1 Vector Addition

If both vectors are the same size of **'n'**, you can combine 2 vectors by adding there **components**.

$$\mathbf{a} + \mathbf{b} = \begin{bmatrix} a_1 \\ a_2 \\ . \\ . \\ . \\ a_n \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ . \\ . \\ . \\ b_n \end{bmatrix} = \begin{bmatrix} a_1 + b_2 \\ a_2 + b_2 \\ . \\ . \\ . \\ a_n + a_b \end{bmatrix} \tag{12}$$

Example:

$$\mathbf{a} + \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} + \begin{bmatrix} 5 \\ 6 \\ 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \\ 10 \\ 12 \end{bmatrix} \tag{13}$$

2.3.2 Vector Subtraction

If both vectors are the same size of **'n'**, you can subtract 2 vectors by adding there **components**.

$$\mathbf{a} - \mathbf{b} = \begin{bmatrix} a_1 \\ a_2 \\ . \\ . \\ . \\ a_n \end{bmatrix} - \begin{bmatrix} b_1 \\ b_2 \\ . \\ . \\ . \\ b_n \end{bmatrix} = \begin{bmatrix} a_1 - b_2 \\ a_2 - b_2 \\ . \\ . \\ . \\ a_n - a_b \end{bmatrix} \tag{14}$$

Example:

$$\mathbf{a} + \mathbf{b} = \begin{bmatrix} 5 \\ 6 \\ 7 \\ 8 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \end{bmatrix} \tag{15}$$

2.3.3 Vector Scalar Multiplication

Vector Scalar Multiplication is the act of multiplying a vector by a Scalar. You multiply the scalar to all **components** inside the vector.

$$\mathbb{C} \cdot \mathbf{a} = \mathbb{C} \cdot \begin{bmatrix} a_1 \\ a_2 \\ . \\ . \\ . \\ a_n \end{bmatrix} = \begin{bmatrix} \mathbb{C} \cdot a_1 \\ \mathbb{C} \cdot a_2 \\ . \\ . \\ . \\ \mathbb{C} \cdot a_n \end{bmatrix} \tag{16}$$

Example:

$$5 \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \\ 15 \\ 20 \end{bmatrix} \tag{17}$$

2.3.4 Dot Product

A dot product (also known as a **inner product** is the sum of all components between to vectors.

Algebraically it can be used to measure the similarity between 2 vectors. If you have 2 vectors of the same **'n'**:

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ . \\ . \\ . \\ u_n \end{bmatrix}, \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ . \\ . \\ . \\ v_n \end{bmatrix} \tag{18}$$

Then the **dot product** would be:

$$\mathbf{u} + \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n \tag{19}$$

Example:

$$\mathbf{u} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 4 \\ -1 \end{bmatrix} \tag{20}$$

$$\mathbf{u} * \mathbf{v} = (2 \cdot 4) + (4 \cdot -1) = 8 - 3 = 5 \tag{21}$$

Geometrically the dot product can also relate to angles between vectors:

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| cos\theta \tag{22}$$

- $\|\mathbf{u}\|$ is the magnitude (length) of $\mathbf{u}$
- θ is the angle between $\|\mathbf{u}\|$ and $\|\mathbf{v}\|$

Useful notes:

- If $\mathbf{u} \cdot \mathbf{v} > 0$: Vectors are pointing in a similar direction.
- If $\mathbf{u} \cdot \mathbf{v} < 0$: vectors are pointing in a opposite directions.
- If $\mathbf{u} \cdot \mathbf{v} = 0$: vectors are perpendicular

2.3.5 Cross Product

Cross product is a operation that creates a new vector perpendicular to the inputted ones. This only works with 3D Vectors.

$$\mathbf{uv} = \begin{bmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{bmatrix} \tag{23}$$

2.3.6 Normalization

Normalization converts the inputted vector into a **unit vector**.

$$\hat{a} = \frac{\mathbf{a}}{||\mathbf{a}||} \tag{24}$$

When a vector is **hatted** (e.g., $\hat{a}, \hat{b}$)